

1. Show that $m(a+bX) = a + b \cdot m(X)$

$$m(X) = \text{mean}$$

$$m(a+bX) = \text{mean of } a+bX$$

$$m(a+bX)$$

$$m(a) + m(bX)$$

mean
of constant

$$a + m(bX)$$

$$a + b \cdot m(X)$$

2. Show that $\text{cov}(X, a+bY) = b \cdot \text{cov}(X, Y)$

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N ((X_i - m(X))(Y_i - m(Y)))$$

$$\text{cov}(X, a+bY)$$

$$\begin{matrix} \uparrow & \uparrow \\ X & Y \end{matrix}$$

$$= m((X - m(X))(a + bY - m(a + bY)))$$

$$= m((X - m(X))(\cancel{a} + bY - (\cancel{a} + b \cdot m(Y))))$$

$$= m((X - m(X))(bY - b \cdot m(Y)))$$

$$= b \cdot m((X - m(X))(Y - m(Y)))$$

$$= b \cdot \text{cov}(X, Y)$$

3. Show that $\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$

and particularly that $\text{cov}(X, X) = S^2$

$$\text{cov}(a+bX, a+bX)$$

$$\begin{matrix} \uparrow & \uparrow \\ X & Y \end{matrix}$$

$$= m((a+bX - m(a+bX))(a+bX - m(a+bX)))$$

$$= m((\cancel{a} + bX - \cancel{a} - b \cdot m(X))(\cancel{a} + bX - \cancel{a} - b \cdot m(X)))$$

$$= m((bX - b \cdot m(X))(bX - b \cdot m(X)))$$

$$= b^2 \cdot m((X - m(X))(X - m(X)))$$

$$= b^2 \text{cov}(X, X)$$

$$S^2 = \text{Var}(X) = m((X - m(X))^2)$$

$$\text{cov}(X, X) = m[(X - m(X))(X - m(X))]$$

$$= m[(X - m(X))^2]$$

$$= S^2$$

4. Non-decreasing transformation

$g()$, is it always true

$$m(g(X)) = g(m(X))?$$

Answer: NO

$$X = 2, 4$$

$$g(X) = X^3$$

$$m(X) =$$

mean of

$$X =$$

$$\frac{2+4}{2} = \frac{6}{2} = 3$$

$$g(m(X))$$

$$g(3) \rightarrow (3)^2 = 27$$

$$m(g(X))$$

$$2^3 = 8$$

$$4^3 = 64$$

$$m(g(X)) = \frac{8+64}{2} = 36$$

$$27 \neq 36$$